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Lightweight Object Detection for African Penguin Population Monitoring

BACKGROUND RECAP

- The African Penguin population in South Africa has declined by about 73% in three decades
- Managers need accurate population data to indicate whether decline continues and whether conservation works.
- Marking birds requires high effort and time (slow) and is a harmful process linked to lower survival rates
- Manual counts are slow and error-prone. Two experts counting the same drone image differed by up to 11.2%
- Published penguin detectors ran on GPUs: a GTX 1070 and 2080. Field systems have run on standard laptops



Sherley et al. (2021), in van den Berg (2025); Geldenhuys (2018); Liu et al. (2020); Tuia et al. (2022); Wu et al. (2023); Sherley et al. (2010); Velasco-Montero et al. (2024); Mohammed (2025)

BACKGROUND RECAP

SINGLE STAGE DETECTORS (YOLO):

- The image is processed once, and boxes and class scores are predicted together as one regression step = fewer computations/ steps in an image = better suited to CPU and edge deployment.
- Two-stage detectors such as Faster R-CNN first propose regions, then classify each one. This is accurate on small, crowded targets but slower to train and run
- On penguins, a YOLO-based detector beat Faster R-CNN by 8.5 mAP
- YOLO comes in nano to extra-large versions of the same design, so the cost of shrinking an architecture can be measured directly



Sherley et al. (2021); in van den Berg (2025); Geldenhuys (2018); Liu et al. (2020); Tuia et al. (2022); Wu et al. (2023); Sherley et al. (2010); Velasco-Montero et al. (2024); Mohammed (2025)



Problem Statement

Published penguin detectors reach near-expert counts, but on GPUs. Liu et al. (2020) came within 2% of expert counts using two GPUs, and Wu et al. (2023) used an NVIDIA V100. How much counting accuracy is lost on a standard laptop CPU has not been reported.

This study tests whether lightweight detectors can **detect and count** penguins on a CPU, staying within 5 points of larger models on mAP@50 and within 5% of the true count, a target stricter than the 11.2% gap between two expert counters (Liu et al., 2020).

Research Objectives

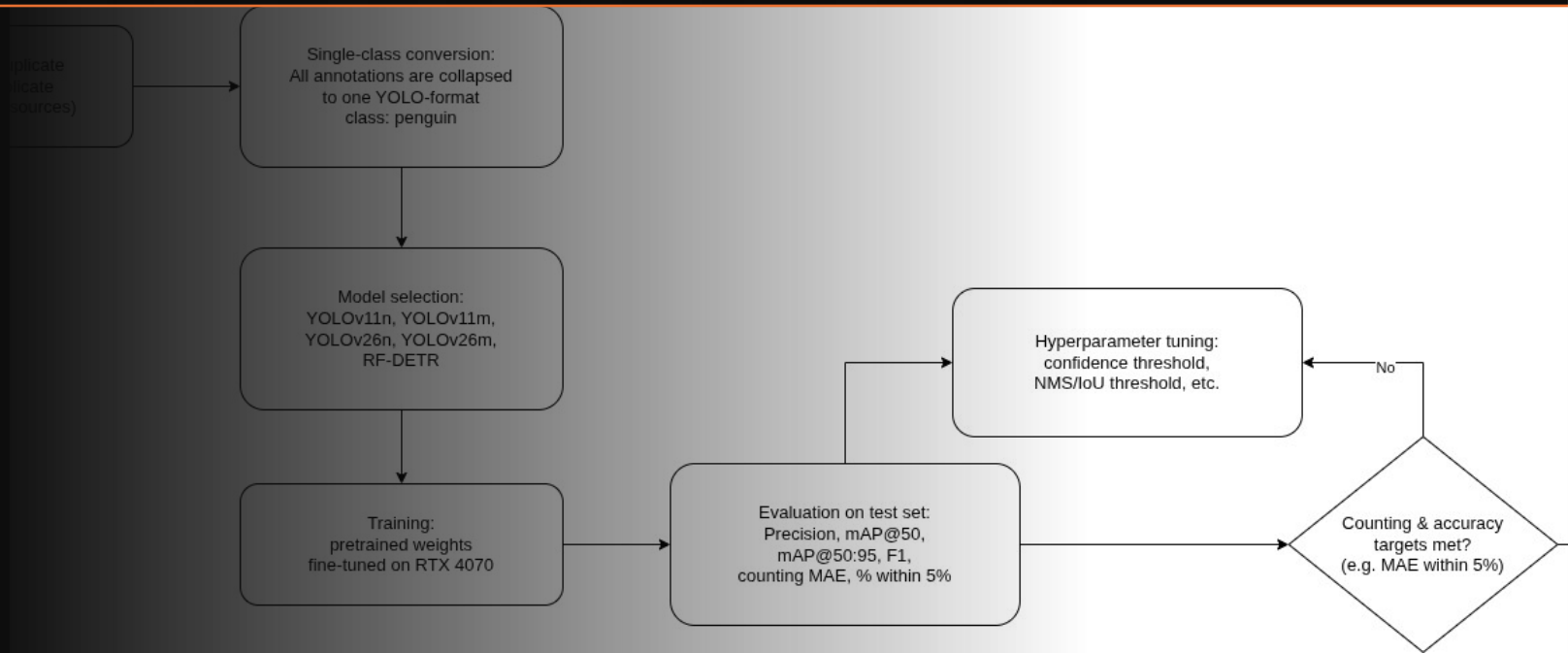
RO1: Determine whether lightweight YOLO models can detect and count African Penguins in field photographs within 5 points of larger models on mAP@50 and within 5\% of the true count.

RO2: Determine whether OpenVINO reduces the CPU latency of these models compared with native PyTorch, and how many images per second they can process on a laptop CPU.

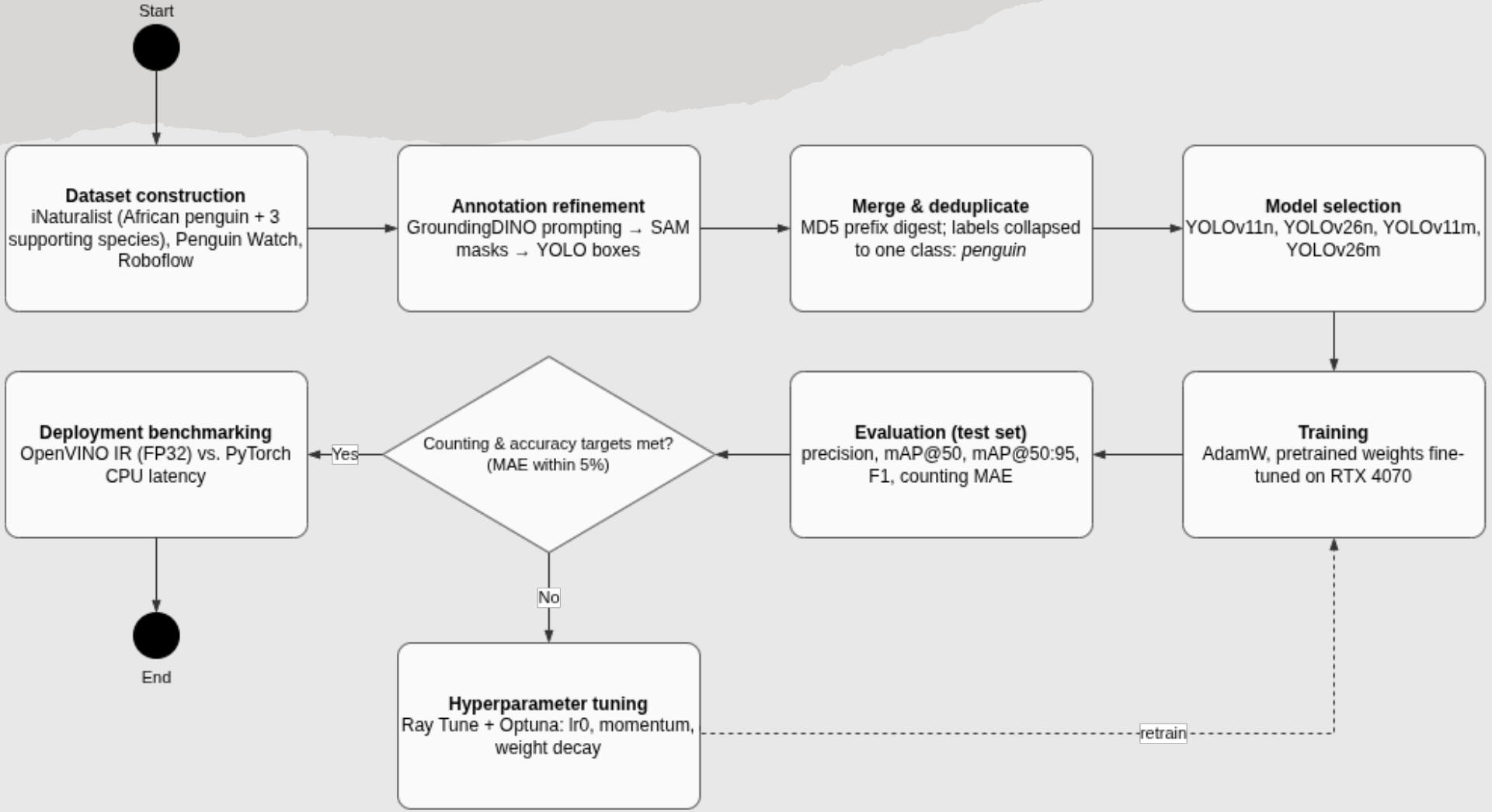
RO3: Quantify how much detection and counting accuracy is lost for the speed gained when moving from medium to nano YOLO models.



PROPOSED MODEL DESIGN OVERVIEW



PROPOSED MODEL DESIGN OVERVIEW / PIPELINE



- **High-level overview:** Dataset acquisition → Preprocessing → Model selection → Training → Evaluation →Deployment benchmarking
- A hyperparameter-tuning loop sits between evaluation and deployment — models are re-evaluated until counting and accuracy targets are met

Preprocessing

1. Sources

(39,416 images before merging, **35,945** *after)*

- 8,661 African Penguin photos (iNaturalist): training, validation, test
- 20,228 photos of three related species: training only
- 3,082 GEORa colony images (Oxford Dataset Penguin Watch): training only
- 7,445 images from 16 Roboflow-hosted sets: training only

SPLIT	PHOTOS	SOURCES
Train	33,357	
Validation	1,733	
Test	855	
	Total = 35 945	

2. Merge

- One class: penguin
- Exact duplicates removed by MD5 and image hashing

3. Split

- Validation and test: African Penguin photos only

Carion et al. (2025); Jones et al. (2020); Arteta et al. (2016); iNaturalist, Penguins of South Africa project

Model Selection & Architecture Overview

Model	Family	Role
YOLOv11n	YOLO	Primary lightweight detector — edge deployment
YOLOv11m	YOLO	Accuracy ceiling within the v11 family
YOLOv26n	YOLO	Lightweight detector — improved small/dense-object handling
YOLOv26m	YOLO	Accuracy ceiling within the v26 family

WHY NANO FOR EDGE DEPLOYMENT?

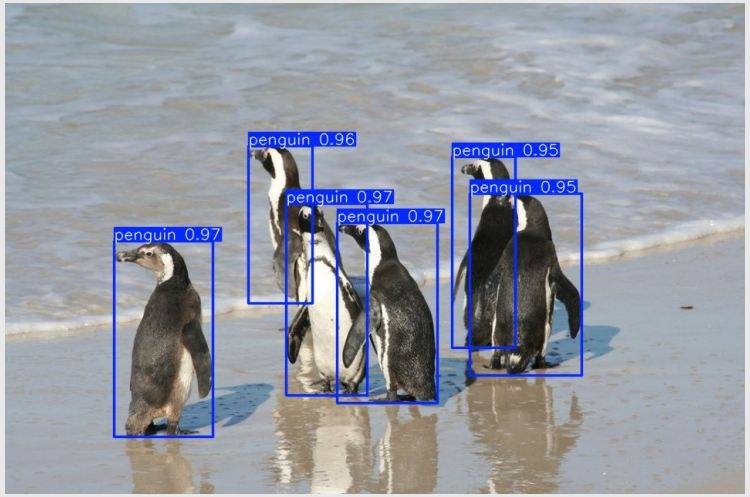
- Deployment constraint: CPU-only field laptops, no dedicated GPU
- Nano variants prioritised for real-time inference under this constraint
- FPS was compared between nano and medium models to measure the speed–accuracy trade-off

Model training

Inference Comparison



Nest scene



Beach scene: 6 Penguins Detected



Occlusion Problem vs Manual Counting

Confidence comparison

Scene	v11m	v26m
Beach (5 penguins)	0.94–0.96	0.95–0.97
Nest-box (occluded scene)	1 detected (0.91)	2 detected (0.94, 0.39)

YOLOv26m: consistently higher confidence, and recovers a partially hidden penguin that YOLOv11m missed entirely.

Experimental Setup

Item	Setting
Data	35, 945 merged images. Test 855 African Penguin iNaturalist Photos
Labels	SAM 3 boxes, checked and corrected in Roboflow
Models	YOLOv11n, YOLOv11m, YOLOv26n , YOLOv26m from COCO-pretrained weights
Training	Up to 100 epochs at 640px on one RTX 4070. Each model trained once
Benchmark	Intel Core Ultra 5 235 CPU. OpenVINO FP32 and native PyTorch
Metrics	mAP@50, mAP@50:95, counting MAE, photos within 5% of the true count, latency per photo

Result 1: Detection Performance

Model	mAP@50	MAP@50:95	Precision	Recall
YOLOv11n	0.8882	0.6648	0.921	0.807
YOLOv26n	0.8838	0.6674	0.919	0.797
YOLOv11m	0.9077	0.7043	0.903	0.841
YOLOv26m	0.9277	0.7440	0.935	0.862

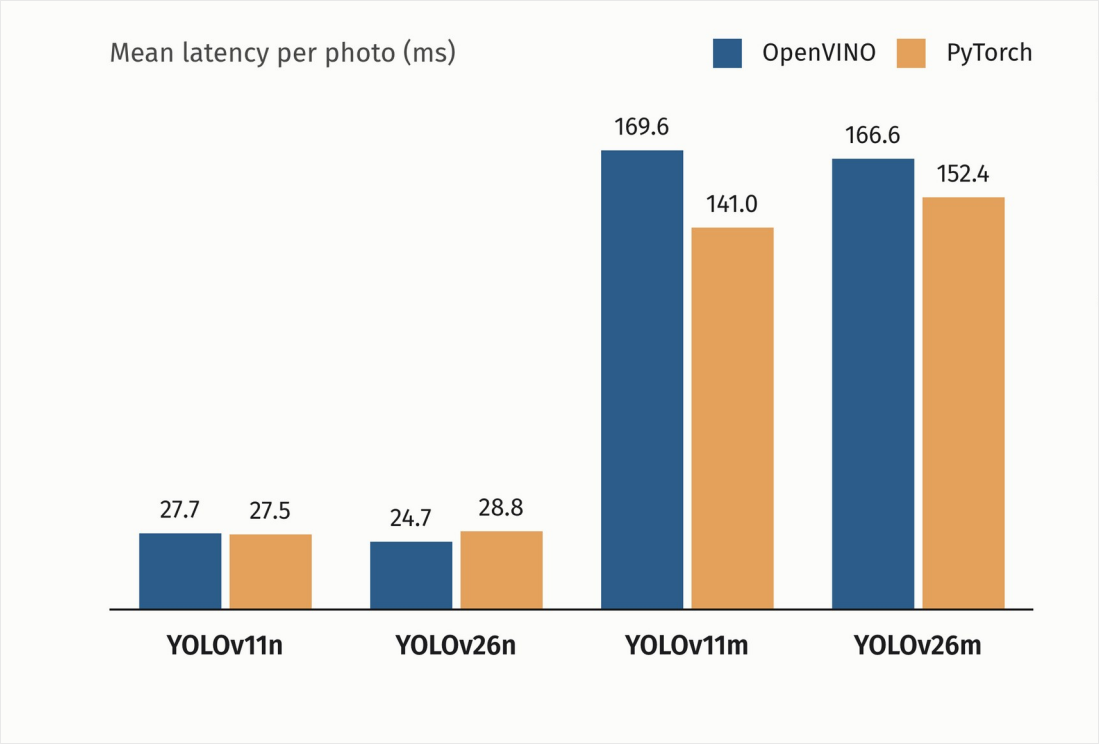
- YOLOv26m was best on every metric: 2.0 points above YOLOv11m on mAP@50 and 4.0 on mAP@50:95
- The larger gain on mAP@50:95 means its boxes were tighter, not just more numerous
- At nano size, the two versions were level: within 0.4 points

Result 2: Counting Accuracy

Model	MAE per photo	Photos within 5%
YOLOv11n	0.88	70.8%
YOLOv26n	0.94	68.2%
YOLOv11m	0.90	67.6%
YOLOv26m	0.72	67.6%

- YOLOv26m had the lowest error, 0.72 penguins per photo. YOLOv11n counted the most photos within 5%

Result 3: CPU Latency, OpenVINO vs PyTorch



- OpenVINO did not halve latency
- The medium models ran faster on PyTorch: 141.0 vs 169.6 ms and 152.4 vs 166.6 ms
- YOLOv26n vs YOLOv11n: 11% faster on OpenVINO, 5% slower on PyTorch. Reported elsewhere: up to 43% faster
- Nano: 35 to 41 photos per second. Medium: 6 to 7

Result 4: Tuning Sensitivity of YOLOv11n

Run	Epochs	Validation mAP@50:95
Default settings (Baseline)	100	0.6815
Default settings at epoch 20	20	0.652
10 Optuna Trials	20 each	0.6706 to 0.6790
Best trial retrained	20	0.6796

- At 20 epochs, the tuned settings scored 1.8 to 2.7 points higher than the defaults
- After 100 epochs, the default run reached 0.6815, level with the tuned retrain
- Tuning sped up convergence but did not raise the final score

Searched learning rate, momentum and weight decay. Trials: AdamW.

Conclusion

Mohammed, S. Y. (2025). Architecture review: Two-stage and one-stage object detection. Franklin Open, 10003 22

Limitations and Future Work

Limitations

- Each model trained once
- Few dense scenes: ~6.7 penguins per test photo
- FP32 only. The integrated GPU was not reachable in OpenVINO

Future work

- Repeat runs with three seeds
- Quantisation: measuring accuracy and speed again
- Knowledge distillation: train the nano model to imitate YOLOv26m, aiming to close the 4-point gap



Timeline

Date	Milestone / Activity
23 Feb 2026	Draft proposal submission
24 Feb – 10 Mar 2026	Seminar Series 1: Proposal presentation and feedback integration
16 Mar 2026	Finalised proposal submission
4 May 2026	Literature Review submission (updated problem statement, questions, objectives)
May 2026	Dataset preparation and bounding box generation script implementation
15 Jun 2026	Begin drafting research design / methodology chapter
VAC (Jun/Jul)	Model implementation and baseline training
7 - 14 Jul 2026	Seminar Series 2: Project implementation presentation
31 Jul 2026	Progress report submission
Aug 2026	OpenVINO optimization and CPU/iGPU benchmarking
Sep 2026	Accuracy–efficiency trade-off analysis and statistical evaluation
18 Sep 2026	Full draft research report submission
2 Oct 2026	Short paper submission (if required)
12–14 Oct 2026	Seminar Series 3: Final results presentation
16 Oct 2026	Final thesis submission
5 Nov 2026	Oral examination
6 Nov 2026	Final corrected submissions (report and ACM short paper)

References

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Questions?

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